# FORM 2

**THE PATENTS ACT, 1970**
**(39 of 1970)**
**&**
**The Patents Rules, 2003**

## COMPLETE SPECIFICATION

**(See section 10; rule 13)**

### 1. TITLE OF THE INVENTION

# "VIRTUAL SENSING SYSTEM FOR WATER QUALITY MONITORING AND METHOD THEREOF"

### 2. APPLICANT (S)

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|---|---|
| (a) NAME: | NATIONAL INSTITUTE OF TECHNOLOGY PUDUCHERRY |
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### 3. PREAMBLE TO THE DESCRIPTION

The following specification particularly describes the invention and the manner in which it is to be performed.

NAME OF THE APPLICANT: NATIONAL INSTITUTE OF TECHNOLOGY PUDUCHERRY

# VIRTUAL SENSING SYSTEM FOR WATER QUALITY MONITORING AND METHOD THEREOF

## FIELD OF THE INVENTION

The present invention relates to the field of aquaculture sensing systems. More particularly the present invention relates to a cost effective virtual sensing system for water quality monitoring (WQM). Further the present invention relates to a method of working of the cost effective virtual sensing system for water quality monitoring (WQM) using less number of sensors, where costly physical sensors are replaced with virtual sensors implemented using solubility equation and inter-dependency relations.

## BACKGROUND OF THE INVENTION

Aquaculture is considered as an integral component of global food security as the world population is expected to reach 9.8 billion by 2050. The harvest from natural resources is mostly beyond their biological limits, and the ocean alone may not meet the demand for food required for this growing population. Aquaculture is one of the efficient solutions for food production but, it has impact on the environment.

Aquaculture practices may involve inefficient energy utilization and wastage of water resources. To minimize the impact on the environment and to optimize the use of resources, Water Quality Monitoring [WQM] plays an important role in aquaculture.

Usually, the farmer assesses the farm by continuous manual monitoring, which will not provide all the required information. This traditional method is tedious, time-consuming, and incapable of meeting the requirements. They can take actions only when the farmer has observed a visible change in the condition of the fish or environment. By this time, environmental factors have already changed dramatically. The procedure to bring the farming environment back to normal is expensive once the damage is already happened.

There are also other traditional methods of water quality monitoring where the water samples are collected and send to a lab. This method consumes time, and they delay the decision-making process affecting productivity.

But the WQM can be done by using sensors that monitor the physical environment continuously provides us with extensive data which may include all the factors which farmer cannot assess manually. This may help us to control the system and reduce the impact on the environment. WQM may also help us with enhancing the quality of the environment, improving product quality, and increasing production. Thus, monitoring of the fish farm optimizes the use of resources improving sustainability and may help the farmer making a better profit.

The vital water quality parameters in aquaculture are temperature, dissolved oxygen [DO], nitrate, ammonia, turbidity, salinity, pH level and alkalinity. The requirement varies for different species, and farmer needs to be aware of them. The fluctuation in these parameters may result in massive production lose. Several reasons are there for contamination of water. Even seawater may be polluted due to urbanization and industrialization.

Dissolved Oxygen [DO] is one of the critical water quality parameters, which reflects changes in water quality in Aquaculture. The water quality of an Aquaculture pond has a direct impact on the growth of aquatic animals and product quality. These parameters directly affect fish health and feed utilization affecting the growth rate. Fish experience stress and diseases breakout when the temperature is chronically near their maximum tolerance or fluctuates suddenly.

The water quality management carried out by farmers is generally based on assumptions and is done periodically without any assessment. In some cases, they will replace the water once in a fixed number of days without considering the

requirements and source may be even more polluted. Some others may use different chemicals periodically. All these practices without proper water quality monitoring will lead to mismanagement and pollution of natural resources. Most of the small scale farmers are not using any monitoring techniques for water quality. A few of them are periodically using different manual testing techniques to check water parameters. One segment of traditional farmers is unaware of technological advances and technology is not affordable for many who are aware of it.

Significant portion of aquaculture production comes from small farms, and helping them to be sustainable as well as efficient will have a considerable impact on the overall production output and reduce the impact of aquaculture on the environment.

The challenge here is to reduce the cost of the system. Systems presently available are mostly not affordable to small scale farms. Moreover, these small-scale farms together contribute to the significant portion of overall production. After a careful study, some parameters are identified, which can be virtually sensed using other parameters. Thus, replacing some of the costly physical sensors. This system works by using the solubility equations for the estimation of the dissolved oxygen in the water. The system is modular, portable, low cost and shares the information with farmers on time, enabling them to manage the farm efficiently. The high price is one reason which prevents small farms from using presently available WQM systems. Reducing the cost of the system will encourage more farmers to use the WQM system. Against the context of the above limitations, a low-cost WQM system is needed that is capable of monitoring and passing information to farmers. Helping them to regulate environmental factors that affect fish wellbeing so that farmers take a more educated and reasoned decision before the outbreak becomes uncontrollable.

The water quality monitoring systems presently available use all the sensors to sense each of the water quality parameter. All these systems generally have in Aquaculture

is evolving towards intensive controlled environment production, giving a substantial improvement in production. Yields rely upon the water quality of the water body. The water quality is determined based on water parameters. These water quality parameters should be monitored regularly for maximizing yields of aquaculture. Moreover, these water parameters differ a lot throughout the day and may quickly change contingent due to any minor change in outside ecological conditions. Therefore, it is vital to monitor these water parameters with high frequency.

Existing WQM systems have a set of sensors, gateways, and computer. Sensors monitor water quality parameters and water level using sensors, effectively reducing the risk of mortality by ensuring continuous water quality monitoring. The system can detect and control temperature, dissolved oxygen content, pH value and water quality parameters in real-time. Most of the WQM systems use ZigBee for communication. Conventional systems use different sensors for water quality parameter sensing of each parameter, making the system expensive.

This work aims to reduce the cost of the water quality monitoring systems used for Aquaculture. Aquaculture plays a vital role in providing food security to the world in the future, and it is increasing steadily as one of the most sustainable methods of food production. The monitoring of water quality has a great significance in aquaculture. The control of water quality in aquaculture optimizes the use of resources, improve sustainability and profitability. The health of the farm is access through monitoring of water quality parameters such as salinity, temperature, pH, and dissolved oxygen. The water quality affects fish feeding behaviour and growth rate. The approach here is to use cost-effective sensors and replace costly sensors by applying Virtual Sensing. Solubility equations and inter-dependency relations for determining the dissolved oxygen in water can replace the expensive physical dissolved oxygen sensor. An extensive study on sensors and interdependency equations helps in identifying low-cost, reliable sensors and reduce the cost of the system. The high price is making

water quality monitoring systems unaffordable for small farms. Using the latest technological advances in open-source hardware and software, we can reduce the cost to such an extent to make the system more affordable. We propose a method to reduce the cost of water quality monitoring system for aquaculture using virtual sensing.

TWI658273B relates to a water quality monitoring system and monitoring method thereof. Disclosed is a water quality monitoring system and a method thereof, comprising: at least one sensor device, a water parameters monitoring device and output data display device, wherein the sensor device is used to sense water to be measured to obtain the sensing raw data of water parameters to be measured. The monitoring device receives the raw data of water parameters measured from the sensor device. The artificial intelligent deep learning technology, the mathematical model and algorithm are embedded to analyze the water parameters, to obtain a plurality of water quality parameters and to increase the accuracy of measured data. The data of multiple water quality parameters are compared with those of preset critical values to judge the water quality state of the measured water. From the data comparisons, the monitoring device evaluates if the water quality is in a dangerous state. In dangerous states, the monitoring device launches a warning sound or/and transmits a warning message to a mobile device or/and initiates a water quality improvement system or/and sends the data to the cloud server for data storage and analysis. In summary, the system provides the monitoring of water quality, the storage of measured data, and the transmission of measured data by e-mail and/or text messaging, the evaluation of abnormal water quality parameters, the pre-warning messages, and the intelligent start of water quality improvement system. In addition, the measured water quality parameters are also sent to the cloud server for storage and analysis. The data output display device could be a mobile device or other data display device. This invention uses fewer sensors to estimate the value of more water quality parameters. Those works have used an artificial neural network to estimate

other water quality parameters without sensors. In our work, we have used the concept of virtual sensing to estimate the value of Dissolved oxygen and salinity. Moreover, this algorithm is implemented at the sensor node in the microcontrollers. We do not require ANN or high complex system for computation to estimate these values.

Applied Water Science (2018) 8:130 relates to a smartphone-based System for water quality analysis. Water quality in rural areas is difficult to monitor due to a lack of connectivity from different water laboratories. In other areas, location-based real-time water quality data collection is tedious and highly dependent on human intervention. The presented paper introduces a low-cost battery operated smartphone-based embedded system design to measure different water quality parameters in various remote locations. Developed system measures pH, total dissolved salt (TDS) and temperature of the water samples using off the shelf available sensors. Measured pH and TDS dataset have been used to derive other water quality parameters using standard mathematical relationships such as salinity, oxygen reduction potential and conductivity. The front-end readout interface circuit has been designed and interfaced with an 8-bit microcontroller along with a classical Bluetooth module for measurement, data acquisition, and logging purpose. A dedicated smartphone-based application offers analysis and cloud data storage possibilities. It also provides the facility to analyze water quality data with location information on Google map for quick judgment and easy understanding. The developed smartphone-based application provides the facility of auto-calibration feature for rapid and on-site usage. The developed smartphone-based application also opens up the possibility to share the data and warnings using different options such as SMS, WhatsApp and E-mail. Overall device has dimensions of 11.0 × 8.0 × 4.0 (in cm), weighs 350 g and runs with 9 Volt rechargeable battery. Obtained results have been validated with a standard water quality measurement system from Eutech Instruments, and it has been

observed that measured and calculated parameters are acceptable according to Indian water quality standards. Various statistical and artificial neural network-based modeling techniques have been used to convert measured water quality parameters to a single water quality index for easy and rapid judgment. The developed water quality measurement system has been used for multiple applications to explore the utility of the system, such as instant water quality judgment and real-time water quality analysis of different water sources. One of the other explored applications is the real-time water quality monitoring of small ponds and lakes. This invention uses fewer sensors to estimate the value of more water quality parameters. Those works have used an artificial neural network to estimate other water quality parameters without sensors. In our work, we have used the concept of virtual sensing to estimate the value of Dissolved oxygen and salinity. Moreover, this algorithm is implemented at the sensor node in the microcontrollers. We do not require ANN or high complex system for computation to estimate these values.

CN105549482A relates to a water environment salinity and dissolved oxygen amount wireless monitoring system. The invention relates to a water environment salinity and dissolved oxygen amount wireless monitoring system. The system comprises sensor nodes, a ZigBee wireless transceiving module, a host computer, a liquid crystal display, a power supply module and a solar power supply module. The sensor nodes communicate with the ZigBee wireless transceiving module, the ZigBee wireless transceiving module is connected with the host computer and the liquid crystal display, and the power supply module supplies power to the ZigBee wireless transceiving module and the liquid crystal display; each sensor node comprises a salinity sensor, a salinity conditioning circuit, a dissolved oxygen amount sensor, a dissolved oxygen amount conditioning circuit and a ZigBee wireless transceiving module, wherein the salinity sensor and the salinity conditioning circuit are connected, the dissolved oxygen amount sensor and the dissolved oxygen amount conditioning circuit are connected, and the salinity conditioning circuit and the

dissolved oxygen amount conditioning circuit are respectively connected with the ZigBee wireless transceiving module; and the solar power supply module supplies power to the sensor nodes. The system provided by the invention has a certain actual application value in factory aquatic product culture production. The system provided by the invention has a certain actual application value in factory aquatic product culture production.

CN101339179B relates to a water quality remote dynamic monitoring system and method for aquiculture. The invention relates to a water quality remote dynamic monitoring system used for aquiculture; the invention adopts the water quality remote dynamic monitoring system to realize real-time dynamic remote wireless monitoring of multiple water quality parameters, which takes the water temperature, the pH value, the electrical conductivity, the dissolved oxygen, the oxidation-reduction potential, the salinity, the turbidity and the illuminance as a priority, can extend to monitoring more water quality parameters simultaneously, and establishes a large-scale and distributed water quality remote dynamic monitoring network in different regions. Compared with other water quality online monitoring systems, the water quality remote dynamic monitoring system has easy expansion of monitoring parameters, higher measurement precision and good compatibility, and the cost thereof is reduced by more than a half compared with the similar product imported from abroad. The invention relates to a water quality remote dynamic monitoring system used for aquaculture. The system can monitor multiple water quality parameters such as pH, dissolved oxygen, salinity, turbidity and illuminance. Radio communication through CDMA communication device and information server.

IN202011017649A relates to a water quality prediction system (WQPS) and method using the internet of things (IoT) and fog computing. A WQPS and method for determining water quality of a water source that includes one or more sensors located at said water source present at one or more geographical locations, the one or more

sensors configured to obtain one or more parameters associated with said water source and transmit said one or more obtained parameters to at least one computing device of a fog networking; and the at least one computing device configured to process said one or more obtained parameters to determine the water quality of said water source located at said one or more geographical locations, and transmit said determined water quality on a display of a user device that triggered determination the water quality of said water source, wherein said one or more geographical locations are different physical locations associated with said water source. A WQPS and method for determining the water quality of a water source with sensors. It includes one or more sensors located at the said water source and other sensors present at different locations. Transmits the parameters to computing device of a fog networking. It displays determined water quality on a display of a user device that triggered determination of the water quality.

Presently available systems use different sensors for sensing the different water quality parameters, making the system expensive. The water quality monitoring has a number of data collection subsystems, which will have all these sensors. These data collection subsystems are deployed in different areas of the water body for collecting data from different parts. The primary cost of the water quality monitoring system is the cost of the sensors. The significant advantages of the proposed scheme, when compared to state of the art, are:

1. This scheme uses virtual sensing to replace costly physical sensors with virtual sensors for reducing the cost of the system.

2. The cost of the sensors will be much higher when implemented in larger areas. Hence by using virtual sensing, we can reduce the cost of the system to a larger extend. Here we replace costly DO sensor and salinity sensor with virtual sensors without compromising on the accuracy.

There exists a need for a cost effective virtual sensing system for water quality monitoring (WQM). Further, there exists a need for a method of working of the cost effective virtual sensing system for water quality monitoring (WQM).

## OBJECTS OF INVENTION

The primary object of the present invention is to provide a low-cost water quality monitoring system that is affordable to small scale farms. And the same time, this system is scalable and can be used in large farms as well.

It is another object of the present invention is to ensure continuous monitoring of the water quality parameters to ensure a safe environment for fish growth and optimizing the use of the resource.

It is another object of the present invention is to increase the quality of aquaculture production along with automation in the pond management system.

It is another object of the present invention, wherein a sustainable solution for aquaculture is proposed.

It is another object of the present invention, wherein the integration of the present solution with a cloud computing platform significantly enhances the aquaculture industry value chain by monitoring the water quality parameters of the pond, impacting fish health.

It is another object of the present invention to provide cloud-based data (water quality) monitor, data storage and data process. It is another object of the present invention to reduce the manual testing and processing cost.

**SUMMARY OF THE INVENTION**

One or more of the problems of the conventional prior art may be overcome by various embodiments of the present invention.

It is primary aspect of the present invention to provide a virtual sensing system for water quality monitoring (WQM), comprising:

one or more nodal units comprises of two or more sensor nodes S, and a central node C; and

a water medium,

wherein the central node C is connected to N number of sensor nodes S, each sensor node S comprises of a transceiver, a microcontroller unit and an antenna with two or more physical sensors and two virtual sensors, the central node C comprises of a transceiver, a microcontroller unit, a display unit, a Global System for Mobile Communications (GSM) modem, an user device, a server and an antenna,

wherein the sensor node S is deployed in the water medium to acquire water quality parameters by physical sensors, and other water quality parameters are virtually sensed using the acquired water quality parameters using physical sensors and the sensed signals are converted to respective water quality parameters by a virtual sensing module in the microcontroller of the sensor node S, the microcontroller of the central node C is provided for controlling and receiving data from the sensor node S, and the value of water quality parameters is virtually sensed from acquired electrical signals by the physical sensors and one or more other parameters are acquired by using the virtually sensed parameters, and

wherein the microcontrollers of the nodal units at the sensor node S is embedded with instructions.

It is another aspect of the present invention, wherein the physical sensors acquire water quality parameters such as temperature, pH, conductivity and the like.

It is another aspect of the present invention, wherein the virtual sensing water quality parameters comprises of dissolved oxygen, salinity and the like.

It is another aspect of the present invention, wherein the microcontroller unit of the nodal units comprises of a memory module and an input/output module and the microcontroller unit also has a virtual sensing module.

It is another aspect of the present invention, wherein the memory module is configured to store the water quality parameters by physical sensors and virtual sensing water quality parameters and the input/output module is connected to the physical sensors configured to receive the sensing signals.

It is another aspect of the present invention, wherein the water quality parameters by physical sensors and virtual sensing water quality parameters are sent to the central node C from the sensor node S using the transceiver and the antenna, received at the central node C and stored in a memory and the central node C communicates the information and uploads the virtual sensing water quality parameters to the server.

It is another aspect of the present invention, wherein the server is a cloud server.

It is another aspect of the present invention, wherein the microcontroller is configured to use the GSM modem to send water quality parameters by physical sensors and virtual sensing water quality parameters information to the user device by the cloud server if online and by the GSM modem if offline.

It is another aspect of the present invention, wherein the user device comprises of one or more of laptop, mobile, smart devices or any other device.

It is another aspect of the present invention to provide a method of working of the virtual sensing system for water quality monitoring (WQM), comprising:

initialization of the physical sensors for virtual sensing water quality parameters in the water medium and output the sensed signal;

reading the physical sensors and storing the values in the memory by the microcontroller;

processing the raw data from each sensor and converts to the virtual sensing water quality parameters values by virtual sensing module of the microcontroller;

virtual sensing of water quality parameters;

sending data to the central node C;

waiting for predefined time;

receiving data from the sensor node S;

storing the data in the memory;

analyzation of the data;

checking whether the parameters are in normal range;

wherein if the parameters are in normal range, the information is communicated to the user and the data is displayed to the user device by the display unit,

wherein if the parameters are in abnormal range, the information is communicated to the user by an alarm unit,

waiting for predefined time;

wherein the operations is to determine to alarm the user or pass information in a defined time interval, and

displaying the virtual sensing water quality parameters values to the user device by the display unit,

wherein the process is repeated continuously passing the information to the end-user and uploading the data to the cloud server.

**BRIEF DESCRIPTION OF THE DRAWINGS**

So that the manner in which the features, advantages and objects of the invention, as well as others which will become apparent, may be understood in more detail, more particular description of the invention briefly summarized above may be had by reference to the embodiment thereof which is illustrated in the appended drawings, which form a part of this specification. It is to be noted, however, that the drawings illustrate only a preferred embodiment of the invention and is therefore not to be considered limiting of the invention's scope as it may admit to other equally effective embodiments.

Fig. 1 illustrates a block diagram of the cost effective virtual sensing system for water quality monitoring (WQM) with central node and 'N' number of sensor nodes according to one embodiment of the present invention.

Fig. 2 illustrates the block diagram of the sensor node of the cost effective virtual sensing system for water quality monitoring (WQM) according to another embodiment of the present invention.

Fig. 3 illustrates the block diagram of the central node of the cost effective virtual sensing system for water quality monitoring (WQM) according to another embodiment of the present invention.

Fig. 4 illustrates the flow chart including the method of sensor initialization, data collection, estimation of parameters from sensed values and communicates the same with the Node C in accordance with the present invention.

Fig. 5 illustrates the flow chart of method of receiving the sensed data at Node C, storing data, analyzing, determination of the result and communicate the result to the end user in accordance with the present invention.

**DESCRIPTION FOR DRAWINGS WITH REFERENCE NUMERALS:**

[100] Virtual sensing system

[110] Water medium

[120] Central node C

[130a.....130n] Sensor nodes S

[210, 310] Microcontroller unit

[220, 320] Transceiver

[230, 330] Antenna

[240a, 240b, 240c] Physical Sensors

[240d, 240e] Virtual sensing water quality parameters

[340] Display unit

[350] Global System for Mobile Communications (GSM) modem

[360] User device

[370] Server

[400a, 400b] Method of working of the virtual sensing system for water quality monitoring (WQM)

[410] Initialization of the sensors for sensing water quality parameters in the water medium and output the sensed signal

[420] Reading the sensors and storing the values in the memory by the microcontroller

[430] Processing the raw data from each sensor and converts to the water quality parameters values by the microcontroller

[440] estimation of water quality parameters

[450] sending data to the central node C

[460] waiting for predefined time

[510] receiving data from the sensor node S

[520] storing the data in the memory

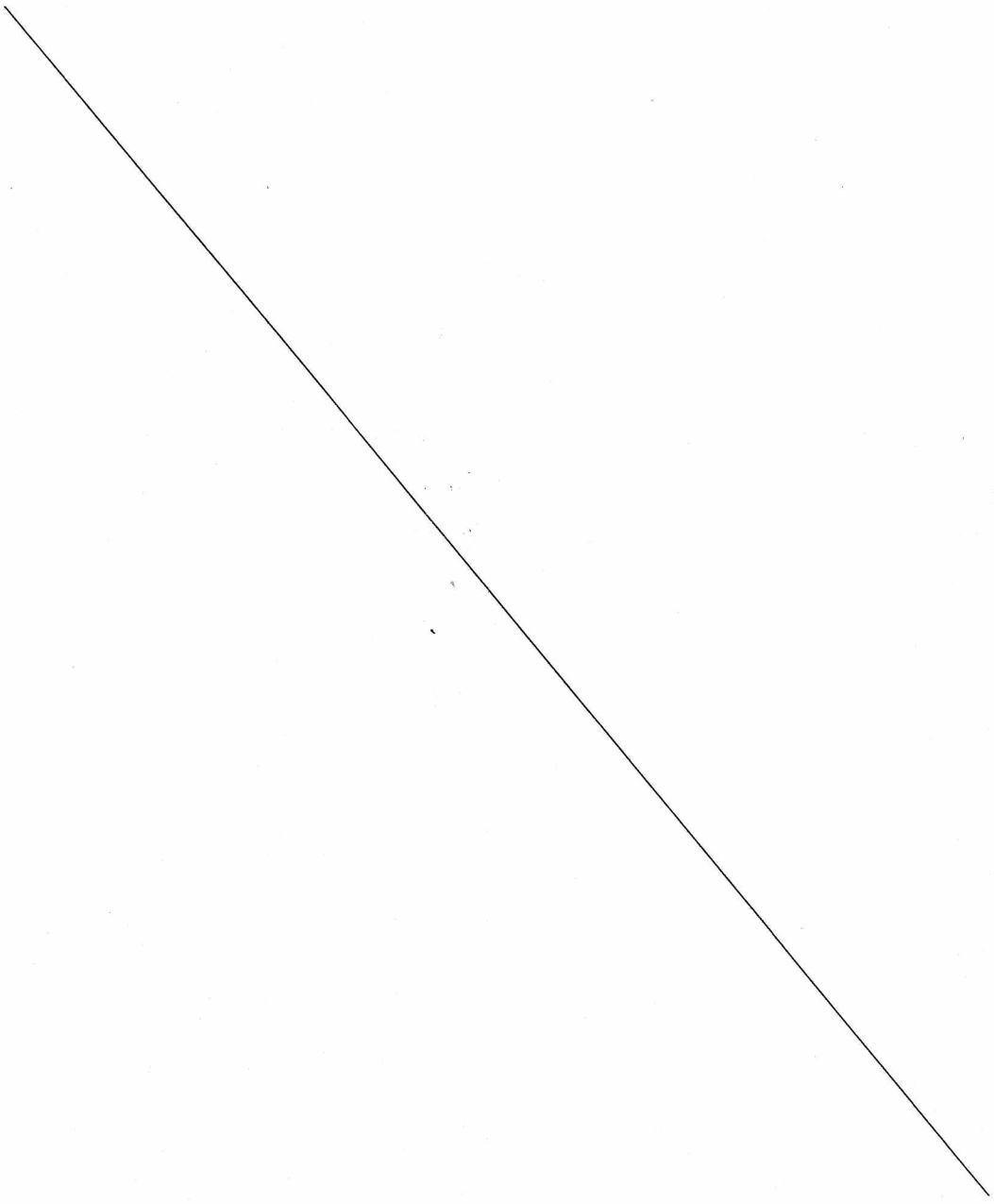
[530] analyzation of the data

[540] checking whether the parameters are in normal range

[550] If the parameters are in the normal range, the information is communicated to the user, and the data is displayed to the user device by the display unit

[560] If the parameters are in the abnormal range, the information is communicated to the user by an alarm unit

[570] Waiting for a predefined time



**DETAILED DESCRIPTION OF THE INVENTION**

It is to be understood that the present disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the drawings. The present disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

The present invention relates to the field of aquaculture sensing systems. More particularly the present invention relates to a cost effective virtual sensing system for water quality monitoring (WQM). Further the present invention relates to a method of working of the cost effective virtual sensing system for water quality monitoring (WQM) using less number of sensors, solubility equation and inter-dependency equations.

The novel aspect of this present system is that it uses Virtual Sensing to replace some physical sensors for monitoring water quality parameters. Replacement of expensive sensors will help in the reduction of the overall cost of the water quality monitoring and reporting system for aquaculture. Aquaculture is an efficient solution for food production but not without impacting the environment. To manage the impact on the environment and improve the efficiency of aquaculture Water Quality Monitoring (WQM) is essential. In this scheme, a novel Virtual Sensing based WQM system for aquaculture is proposed, rather than using separate sensors for each water quality parameters we use virtual sensing to remove some costly sensor reducing the overall cost of the system.

The invention pertains to a low-cost method and an apparatus for monitoring water quality parameters for aquaculture that comprises a central node, a plurality of sensor nodes and embedded system. The sensors sense water quality parameters and

separately output a plurality of sensing signals, including the water quality parameters. The microcontroller system includes a memory module, a virtual sensing module, an input/output module, and a transceiver module. The sensors are connected to the input/output for receiving the sensed signals. The microcontroller converts these sensed electrical signals to corresponding water quality parameters. The memory module stores a plurality of sensor values to the respective water quality parameters. The virtual sensing module in the processor estimates two more parameters without using corresponding sensors but using the sensed parameters. After estimating all the parameters, the data is sent to the central node by the microcontroller. The central node collects this data stores them in the database, pass this information to the user in defined intervals. The central node also determines whether the parameters are in abnormal ranges and issues warning signal according to the determined result. The water quality monitoring system for aquaculture consists of Node C for controlling and receiving the data from many Node S, which have the sensors for sensing water quality parameters. The water quality monitoring system includes a set of sensors and a microcontroller system. The sensors are used to sense the water quality parameters and output a set of sensing signals that correspond to the water quality parameters. The microcontroller system includes a memory module, a virtual sensing module and an input/output module. The memory module is used to store the water quality parameters. The virtual sensing module estimates the salinity using the signal from the conductivity sensor. And dissolved oxygen is also estimated by the virtual sensing module from sensed water quality parameters temperature and estimated salinity. The input/output module connected to the sensors is used to receive the sensing signals. The microcontroller converts all sensed signal to respective water quality parameters and with help of virtual sensing module estimates other parameters without sensors. The water quality parameters are sent to Node C from Node S using the transceiver and antenna, received at Node C and stored in its memory. Further data analysis is done at Node C. Node C communicates the information and uploads the water quality parameters to the cloud.

Fig. 1 shows the water quality monitoring system with the Central Node C [120] and 'N' number of sensor Node S. The sensor Node S [130] may be deployed around the central Node C [120] and may choose any mode of communication with the central Node C. Many sensor Node S [130] is deployed in the waterbody/ water medium [110] to acquire water quality parameters from different areas of the water body/ water medium [110]. The aquaculture water quality monitoring system consists of a central Node C as center point and a set of sensor Node S [130a-n] for sensing water quality parameters and reporting back to central Node C.

The virtual sensing system [100] for water quality monitoring (WQM), comprises one or more nodal units comprises of two or more sensor nodes S [130], and a central node C [120], and a water medium [110]. The central node C [120] is connected to N number of sensor nodes S [130]. Each sensor node S [130a.....130n] comprises of a transceiver [220], a microcontroller unit [210] and an antenna [230] with two or more sensors [240a, 240b, 240c]. The central node C [120] comprises of a transceiver [320], a microcontroller unit [310], a display unit [340], a Global System for Mobile Communications (GSM) modem [350], a user device [360], a server [370] and an antenna [330]. The sensor node S [130] is deployed in the water medium [110] to acquire water quality parameters [240a, 240b, 240c], and the estimation of water quality parameter values [240d, 240e] and environmental parameter values is calculated and the sensed signal is converted to respective water quality parameters [240d, 240e] by the microcontroller [210] of the sensor node S [130]. The microcontroller [310] of the central node C [120] is provided for controlling and receiving data from the sensor node S [130], and the value of water quality parameter values [240d, 240e] is virtually sensed from acquired electrical signals by the physical sensors [240a, 240b, 240c] and one or more parameters by using the virtually sensed parameters [240d, 240e]. The microcontrollers [210, 310] of the nodal units [130, 120] at the sensor node S is embedded with instructions. The sensors acquire environmental parameters such as temperature [240a], pH [240b],

conductivity [240c] and the like. The water quality parameters comprises of dissolved oxygen [240d], salinity [240e] and the like. The microcontroller unit [210, 310] of the nodal units [130, 120] comprises of a memory module and an input/output module. The microcontroller unit [210] also has an additional virtual sensing module. The memory module is configured to store the water quality parameters [240a, 240b, 240c, 240d, 240e], the input/output module is connected to the sensors [240a, 240b, 240c] configured to receive the sensing signals, and the virtual sensing unit estimates the water quality parameters without using physical sensors. The water quality parameters [240a, 240b, 240c, 240d, 240e] are sent to the central node C [120] from the sensor node S [130] using the transceiver [220, 320] and the antenna [230, 330], received at the central node C [120] and stored in a memory and the central node C [120] communicates the information and uploads the water quality parameters [240a, 240b, 240c, 240d, 240e] to the server [310]. The server [310] is a cloud server. The microcontroller is configured to use the GSM modem [350] to send water quality parameters [240a, 240b, 240c, 240d, 240e] information to the user device [360] by the cloud server [310] if online and by the GSM modem [350] if offline. The user device [360] comprises of one or more of laptop, mobile, smart devices or any other device.

Fig. 2 shows various parts of central Node S [120] that is used in the water quality monitoring system [100]. The central Node S [120] includes a microcontroller [210], a transceiver [220], and an antenna [230] with the set of sensors [240a-240c]. Sensors [240a-240c] acquire environmental parameters such as temperature [240a], pH [240b] and conductivity [240c]. The sensor Node S also produces two more water quality parameters without using sensors. The water quality parameters dissolved oxygen [240d] and salinity [240e] is virtually sensed using the parameters already acquired [240a-240c]. The microcontroller [210] includes a memory module, virtual sensing module and an input/output module. The input/output module is connected to sensors [240a-240c] and is configured to receive a plurality of sensing signals that correspond

to the water quality parameters. The microcontroller processes the signal received from sensors through the input/output module and converts available raw signals to water quality parameters values with help of virtual sensing module. Estimation of parameters like DO and salinity is done from temperature and conductivity values using Weiss equation and interdependency equations by the processing of virtual sensing module in microcontroller. All the calculations on raw sensor data are done in microcontroller and the processed data is passed to Node C as water quality parameter.

Fig. 3 illustrates various components in central Node C [130] within the water quality monitoring system [100]. The central Node C [130] includes a microcontroller [310], a transceiver [320] and an antenna [330]. The transceiver receives the data send by sensor Node S through the antenna. The microcontroller depending on the parameter value warns the user and displays the water quality parameters in the LCD [340]. The end-user has the provision to monitor the farm using smartphone [360] by accessing the cloud [370] or communicating with the microcontroller. The microcontroller has the provision to upload the water quality parameters to the cloud and inform the end through the user smartphone. The microcontroller may also use the GSM modem [350] to send information to the smartphone without the use of cloud as well.

Fig. 4 illustrates a flowchart to initialize and sense the water quality parameters in the water body using the set of sensors and output the sensed signal [410]. The microcontroller reads the sensors and stores the values in the memory [420]. The microcontroller processes the raw data from each sensor and converts it to water quality parameters values [430]. The processing for estimation [440] of water quality parameters dissolved oxygen and salinity is done from other parameters temperature, conductivity values using equations [1-4] by the virtual sensing module in the microcontroller. Equation 4 is used for processing the estimation of salinity from conductance by the microcontroller.

$S = [5.572*10^4\ [SC] + 2.03*10^{-9}\ [SC]^2] /1000$ [1]

where S is the salinity in ppt, and T is the water temperature in Kelvin.

Equation 2 is used for processing the estimation of dissolved oxygen from temperature by the microcontroller [210].

$DO = DO_0 * F_S$ [2]

Where DO is the dissolved oxygen in the water body [110], $DO_0$ is the Dissolved oxygen concentration at zero salinity and one-atmosphere pressure in the water body [110] given by equation 3. $F_S$ is the salinity factor given in equation 4.

$$DO_0 = 1.42905\ exp[-173.4292 + 249.6339\ [\frac{100}{T}] + 143.3483\ [In\frac{T}{100}] - 21.8492[\frac{T}{100}]]$$ [3]

where T is the water temperature in Kelvin and DO is the dissolved oxygen in water mg/l.

$$F_S = exp\left[S\left[-0.033096 + 0.14259\left(\frac{T}{100}\right) - 0.0017\left(\frac{T}{100}\right)^2\right]\right]$$ [4]

where S is the salinity in ppt, and T is the water temperature in Kelvin.

The estimation based on equations is done in Node S and the data is sent to Node C as a water quality parameter. The microcontroller waits for the defined time to initialize the sensors and repeat the process. This scheduling mechanism done in sensor Node S [120] helps the system to be energy efficient.

Fig. 5 illustrates the flowchart to receive the data [510] from Node S, this data is stored in the database [520] and the data is analyzed [530]. The operations [540] are to determine to alarm the user or pass information in a defined time interval [560]. The system waits for a specified time in case of parameters are in the normal range [540]. And the process is repeated continuously passing the information to the end-user and uploading the data to the cloud [560].

The method [400a, 400b] of working of the virtual sensing system [100] for water quality monitoring (WQM), comprising the following steps: initialization [410] of the sensors [240a, 240b, 240c] for sensing water quality parameters, and virtual sensing of water quality parameters [240d, 240e] in the water medium [110] and output the sensed signal; reading [420] the sensors [240a, 240b, 240c] and storing the values in the memory by the microcontroller [210]; processing [430] the raw data from each sensor and converts to the water quality parameters values [240d, 240e] by the virtual sensing module in microcontroller [210]; estimation [440] of water quality parameters [240d, 240e]; sending [450] data to the central node C [120]; waiting [460] for predefined time; receiving [510] data from the sensor node S [130]; storing [520] the data in the memory; analyzation [530] of the data; checking [540] whether the parameters are in normal range; if the parameters are in normal range, the information is communicated [550] to the user and the data is displayed to the user device [360] by the display unit [340], if the parameters are in abnormal range, the information is communicated [560] to the user by an alarm unit, waiting [570] for predefined time; the operations [540] is to determine to alarm the user or pass information in a defined time interval [560], and displaying the water quality parameters values [240d, 240e] to the user device [360] by the display unit [340]. The process is repeated continuously passing the information to the end-user and uploading the data to the cloud server [370].

Although the invention has been described and illustrated with respect to the exemplary embodiments thereof, it should be understood by those skilled in the art that the foregoing and various other changes, omissions and additions may be made therein and thereto, without parting from the spirit and scope of the present invention.

NAME OF THE APPLICANT: NATIONAL INSTITUTE OF TECHNOLOGY PUDUCHERRY

**WE CLAIM:**

1. A virtual sensing system [100] for water quality monitoring (WQM), comprising:

one or more nodal units comprises of two or more sensor nodes S [130], and a central node C [120]; and

a water medium [110],

characterized in that, the central node C [120] is connected to N number of sensor nodes S [130], each sensor node S [130a.....130n] comprises of a transceiver [220], a microcontroller unit [210] and an antenna [230] with two or more physical sensors [240a, 240b, 240c] and two virtual sensors, the central node C [120] comprises of a transceiver [320], a microcontroller unit [310], a display unit [340], a Global System for Mobile Communications (GSM) modem [350], an user device [360], a server [370] and an antenna [330],

wherein the sensor node S [130] is deployed in the water medium [110] to acquire water quality parameters by physical sensors [240a, 240b, 240c], and other water quality parameters [240d, 240e] are virtually sensed using the acquired water quality parameters using physical sensors [240a, 240b, 240c], the sensed signals are converted to respective water quality parameters [240d, 240e] by a virtual sensing module in the microcontroller [210] of the sensor node S [130], the microcontroller [310] of the central node C [120] is provided for controlling and receiving data from the sensor node S [130], and the value of water quality parameters [240d, 240e] is virtually sensed from acquired electrical signals by the physical sensors [240a, 240b, 240c] and one or more others parameters are acquired by using the virtually sensed parameters [240d, 240e], and

wherein the microcontrollers [210, 310] of the nodal units [130, 120] at the sensor node S is embedded with instructions.

2. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the physical sensors [240a, 240b, 240c] acquire water quality parameters such as temperature, pH, conductivity and the like.

3. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the virtual sensing water quality parameters comprises of dissolved oxygen [240d], salinity [240e] and the like.

4. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the microcontroller unit [210, 310] of the nodal units [130, 120] comprises of a memory module and an input/output module and the microcontroller unit [210] also has a virtual sensing module.

5. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the memory module is configured to store the water quality parameters by physical sensors [240a, 240b, 240c] and virtual sensing water quality parameters [240d, 240e], and the input/output module is connected to the physical sensors [240a, 240b, 240c] configured to receive the sensing signals.

6. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the water quality parameters by physical sensors [240a, 240b, 240c] and virtual sensing water quality parameters [240d, 240e] are sent to the central node C [120] from the sensor node S [130] using the transceiver [220, 320] and the antenna [230, 330], received at the central node C [120] and stored in a memory and the central node C [120] communicates the information and uploads the virtual sensing water quality parameters [240d, 240e] to the server [310].

7. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the server [310] is a cloud server.

8. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the microcontroller is configured to use the GSM modem [350] to send water quality parameters by physical sensors [240a, 240b, 240c] and virtual sensing water quality parameters [240d, 240e] information to the user device [360] by the cloud server [310] if online and by the GSM modem [350] if offline.

9. The virtual sensing system [100] for water quality monitoring (WQM) as claimed in claim 1, wherein the user device [360] comprises of one or more of laptop, mobile, smart devices or any other device.

10. A method [400a, 400b] of working of the virtual sensing system [100] for water quality monitoring (WQM), comprising:

initialization [410] of the physical sensors [240a, 240b, 240c] for virtual sensing water quality parameters [240d, 240e] in the water medium [110] and output the sensed signal;

reading [420] the physical sensors [240a, 240b, 240c] and storing the values in the memory by the microcontroller [210];

processing [430] the raw data from each sensor and converts to the virtual sensing water quality parameters values [240d, 240e] by virtual sensing module of the microcontroller [210];

virtual sensing [440] of water quality parameters [240d, 240e];

sending [450] data to the central node C [120];

waiting [460] for predefined time;

receiving [510] data from the sensor node S [130];

storing [520] the data in the memory;

analyzation [530] of the data;

checking [540] whether the parameters are in normal range;

wherein if the parameters are in normal range, the information is communicated [550] to the user and the data is displayed to the user device [360] by the display unit [340],

wherein if the parameters are in abnormal range, the information is communicated [560] to the user by an alarm unit,

waiting [570] for predefined time;

wherein the operations [540] is to determine to alarm the user or pass information in a defined time interval [560], and

displaying the virtual sensing water quality parameters values [240d, 240e] to the user device [360] by the display unit [340],

wherein the process is repeated continuously passing the information to the end-user and uploading the data to the cloud server [370].

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